

DiffMark: Diffusion-Based Robust Watermarking Against Deepfake Manipulation

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Abstract—The rapid proliferation of deepfake technology poses significant threats to digital media integrity, enabling malicious manipulation of facial images with near-indistinguishable results. Existing watermarking methods often fail to maintain robustness after deepfake operations such as face-swapping and expression transfer, as these manipulations substantially alter the spatial and frequency characteristics of images. In this paper, we present DiffMark, a diffusion-based watermarking framework specifically designed to withstand deepfake attacks. DiffMark leverages a U-Net-based encoder-decoder architecture tightly coupled with a guided diffusion process to embed imperceptible binary watermark messages into facial images. A key innovation is the use of multiple deepfake noise layers during training, including SimSwap, StarGAN, UniFace, CSCS, and FSRT, enabling the model to learn robust watermark representations that survive diverse manipulation pipelines. We further introduce a guidance mechanism that directs the diffusion sampling process toward enhanced message fidelity under deepfake perturbation. Extensive experiments on CelebA-HQ and LFW datasets at resolutions of 128×128 and 256×256 demonstrate that DiffMark achieves state-of-the-art bit accuracy and perceptual quality, outperforming prior watermarking methods in both robustness and invisibility metrics.

Index Terms—deepfake detection, digital watermarking, diffusion models, robust watermarking, face manipulation, information hiding

I. INTRODUCTION

The emergence of Generative Adversarial Networks (GANs) and latent diffusion models has dramatically lowered the barrier to creating convincing deepfake content. Face-swapping systems such as SimSwap [8] and expression-transfer methods such as FSRT [12] can produce facial images that are visually indistinguishable from authentic photographs, creating serious risks for misinformation, identity fraud, and non-consensual media generation.

Digital watermarking offers a proactive defence strategy: by embedding an imperceptible ownership or provenance signal into an image before distribution, one can later verify whether the image has been manipulated. However, deepfake operations constitute a particularly challenging class of distortions because they jointly modify spatial structures, textures, and semantic content. Classical watermarking methods, designed to withstand compression, resizing, or JPEG artefacts, cannot reliably survive face-swapping pipelines that replace large facial regions.

Why diffusion-based embedding is inherently more robust than GAN-based watermarking. A GAN-based encoder embeds a message in a single forward pass, mapping the host image and message jointly to a watermarked image through one deterministic function. Because this mapping is one-shot, the watermark signal tends to concentrate in whichever spatial-frequency band the encoder network finds most convenient to modify, and any transformation that perturbs that particular band, such as a face-swap network replacing an entire facial region, can remove most of the embedded signal in one operation. A diffusion-based encoder, in contrast, embeds the message incrementally across a multi-step denoising trajectory, injecting the message embedding into every residual block at every timestep via adaptive group normalisation. This has two consequences for robustness. First, the watermark is distributed across multiple noise scales and spatial resolutions rather than concentrated in one, so a manipulation that damages high-frequency detail still leaves lower-frequency, coarser-scale traces of the message intact, and vice versa. Second, because the diffusion process is itself a model of how images can be perturbed and reconstructed, the encoder implicitly learns a representation that is already compatible with reconstruction-style edits, which is analogous to the kind of edit a face-swap or reenactment network performs on a local region. GAN-based encoders have no equivalent inductive bias and must instead rely entirely on the differentiable noise layers seen during training to discover robust features. We stress that this is a structural argument for why diffusion embedding is a good fit for the problem, not a claim that diffusion watermarks are unconditionally unbreakable. The limitations discussion later in this paper addresses an important counter-example, diffusion-based regeneration attacks, where the same generative flexibility that helps embedding robustness can also be turned into an attack that erases the mark.

Recent learning-based watermarking approaches [1]–[4] have significantly improved robustness by training encoder-decoder networks end-to-end against differentiable noise layers. Yet none of these methods explicitly models the complex, non-linear distortions introduced by modern deepfake generators, limiting their effectiveness in practice.

In this work we propose DiffMark, which addresses this gap through three contributions:

- A diffusion-process-based embedding architecture that integrates watermark messages directly into the denoising trajectory of a U-Net, producing watermarked images that are inherently robust to regeneration-based attacks.
- A multi-deepfake noise layer training strategy that simultaneously exposes the watermark encoder to five distinct face-manipulation systems, encouraging the learning of universally robust feature representations.
- A message-guided sampling mechanism that steers the reverse diffusion chain toward high message-extraction accuracy even when the watermarked image has been processed by an unseen deepfake system.

Experiments confirm that DiffMark substantially outperforms prior art in bit accuracy after deepfake attack while maintaining imperceptibility ($\text{PSNR} > 35 \text{ dB}$, $\text{SSIM} > 0.97$) across both tested datasets and resolutions.

II. RELATED WORK

A. Traditional Watermarking

Early watermarking techniques operated in transform domains such as DCT [15] and DWT, encoding bits into frequency coefficients. While resilient to common signal-processing distortions, they are unable to tolerate semantic-level editing, including face swapping.

B. Learning-Based Watermarking

HiDDeN [1] introduced the first end-to-end trainable encoder-decoder framework, pairing the encoder with a differentiable noise layer during training. StegaStamp [2] extended this to physical-world scenarios using spatial transformer networks. SepMark [3] disentangles robust and fragile watermark components, while LampMark [4] employs localized attention pooling to improve spatial fidelity. None of these methods directly targets deepfake distortions.

C. Deepfake Detection and Provenance

Passive deepfake detectors [14] analyse artefacts in image statistics to classify images as real or fake. Active methods embed a signal at capture time so that downstream forgery is detectable regardless of detector evolution. DiffMark is an active method: it does not classify deepfakes but instead enables ownership verification and tampering localisation.

D. Diffusion Models

Denoising Diffusion Probabilistic Models (DDPMs) [5] learn to reverse a Markov noising chain. Guided diffusion [7] introduced classifier guidance to steer generation toward desired attributes. DiffMark repurposes this guidance mechanism to maximise watermark message recovery rather than class probability.

E. Recent Diffusion-Based and Provenance Watermarking (2024–2026)

A growing body of work has moved robust watermarking toward diffusion architectures and, in parallel, toward system-level provenance standards. Zhang et al. [16] embed watermark

bits directly into a pretrained Stable Diffusion backbone rather than a bespoke encoder, trading some training flexibility for the ability to reuse a strong pretrained prior. Wang et al. [17] take a complementary route to ours, using facial-landmark structure to build a perceptual watermark tailored to face-swap forensics rather than a generic deepfake noise layer. Tan et al. [18] propose WaterDiff, which repurposes a pretrained diffusion autoencoder for reversible image-to-latent mapping and embeds the message via a wavelet-domain transform of the latent; unlike DiffMark, its reliance on an invertible mapping tends to trade away fine image detail, whereas our residual-based embedding (Eq. 2) is designed to preserve it. Wen et al. [19] introduced Tree-Ring watermarks, which fingerprint the initial noise pattern of a diffusion sampler rather than the output image, giving strong robustness for content that is itself diffusion-generated but which does not directly apply to watermarking an already-captured photograph, as DiffMark does. Zhang et al. [20] and its 2025 successor OmniGuard [21] extend versatile watermarking to joint tamper-localisation and copyright protection, which is a complementary goal to the pure bit-accuracy objective studied here. Lu et al. [22] present VINE, a diffusion-encoder watermark evaluated at ICLR 2025 that is explicitly benchmarked for robustness to generative image editing; together with an updated TrustMark [23], these represent the current state of the art in generic (non-deepfake-specific) learned watermarking against which DiffMark could usefully be benchmarked in future work (Section VII).

On the provenance side, the Coalition for Content Provenance and Authenticity finalised C2PA specification 2.2 in 2025 [25], which complements pixel-level watermarking with cryptographically signed manifests, and Google’s SynthID-Image [26] deployed learned watermarking of generated images at internet scale. These standards address a different threat model from DiffMark: they focus on labelling AI-generated content and platform-level metadata, whereas DiffMark focuses on surviving an active deepfake attack on an already-watermarked photograph. The two approaches are complementary, however, and could in principle be combined, with a DiffMark-style pixel watermark serving as the soft-binding signal referenced in the C2PA specification.

We also note a concurrent and independently developed system that shares our method’s name and general approach: Sun et al. [27], first released in mid-2025, also propose a diffusion-conditioned watermark trained against a deepfake noise layer. Their design uses a frozen autoencoder to simulate deepfake manipulation during training and a related deepfake-resistant guidance term, whereas the present work trains directly against five named deepfake systems plus an LDM-autoencoder proxy and an “ordinary” distortion set, and uses a message-conditioned adaptive group-normalisation U-Net rather than a frozen autoencoder. We flag the overlap explicitly for the reader’s benefit and view the convergence of two independent groups on similar architectural choices as supporting evidence that diffusion-conditioned embedding is a promising direction for deepfake-robust watermarking, rather than as a conflict between the two papers.

III. LITERATURE REVIEW AND RESEARCH GAP ANALYSIS

Section II surveyed individual prior works; this section synthesises that body of work thematically and identifies the specific gap that motivates DiffMark’s design.

The watermarking literature has progressed through three broad generations. The *first generation* (transform-domain methods such as DCT- and DWT-based spread-spectrum watermarking [15]) offers mathematically well-understood robustness guarantees against signal-processing operations, but was never designed to survive semantic, content-altering edits, and consequently collapses under any transformation that touches the underlying facial identity. The *second generation* (end-to-end learned encoder-decoder frameworks such as HiD-DeN [1], StegaStamp [2], SepMark [3], and LampMark [4]) replaced hand-designed transforms with differentiable noise layers and learned features, substantially improving robustness to common distortions such as JPEG compression, cropping, and blurring. However, as Table I summarises, none of these methods trains against, or evaluates on, deepfake-specific manipulations, so their robustness to face-swapping and reenactment is, at best, incidental. The *third generation*, diffusion-based and provenance-oriented approaches [16]–[23], [25]–[27], is the most recent and the most directly relevant to this work. Within this generation, three distinct sub-strategies are visible: (i) repurposing a pretrained diffusion backbone for watermark embedding [16], [18], (ii) fingerprinting the diffusion sampling process itself rather than the output pixels [19], and (iii) building generic, non-deepfake-specific robustness against arbitrary generative edits [22], [23]. A parallel, complementary track pursues system-level provenance through cryptographic manifests and platform-scale watermarking [25], [26] rather than pixel-level robustness at all.

Taken together, the literature reveals a consistent gap: methods that are robust to conventional signal-processing distortions are not trained or evaluated against deepfake manipulation, while methods that do target diffusion-based robustness either operate on already-generated content (Tree-Ring [19]), embed within a diffusion backbone without deepfake-specific supervision [16], [18], or pursue generic, non-deepfake robustness [22], [23]. Only Wang et al. [17] and the concurrent work of Sun et al. [27] directly target deepfake robustness with a diffusion-adjacent design, and neither trains against as broad a set of named, heterogeneous face-manipulation systems as the five used here. DiffMark is positioned to close this gap by combining a diffusion-native embedding architecture with supervision from five distinct deepfake systems plus an autoencoder proxy, and by adding a message-guided sampling mechanism that specifically targets bit-accuracy recovery rather than tamper localisation or generic edit robustness.

IV. PROPOSED METHOD: DIFFMARK

A. Overview

Fig. 1 shows the overall pipeline. Given a host facial image $I \in \mathbb{R}^{H \times W \times 3}$ and a binary watermark message $m \in \{0, 1\}^L$,

DiffMark produces a watermarked image $\hat{I}$ perceptually similar to I . A decoder network D later extracts $\hat{m}$ from a (possibly deepfaked) version of $\hat{I}$, and bit accuracy is measured as

$$BA = \frac{1}{L} \sum_{i=1}^L \mathbb{1}[\hat{m}_i = m_i]. \quad (1)$$

B. Encoder-Decoder Architecture

The backbone is a guided-diffusion-style U-Net [7] extended with message conditioning. Given message m , we project it to an embedding vector $e_m \in \mathbb{R}^d$ and inject it into every residual block via adaptive group normalisation. The encoder produces a residual ΔI , and the watermarked image is:

$$\hat{I} = I + \tanh(\Delta I) \cdot \varepsilon, \quad (2)$$

where ε is a small scalar controlling perturbation magnitude. The decoder mirrors the encoder with attention pooling at the bottleneck to aggregate spatial features into message logits $\hat{m} \in [0, 1]^L$. Table II summarises the key architectural hyperparameters.

C. Diffusion-Based Embedding

DiffMark treats watermark embedding as a conditioned denoising problem. The forward process is:

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad (3)$$

where $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ and $\{\beta_t\}$ follows a linear schedule over $T = 100$ steps. The model learns the reverse process $p_\theta(x_{t-1} | x_t, m)$ by predicting x_0 directly. At inference, DDIM [6] with 10 steps (ddim10) is used for fast, high-quality embedding.

Algorithm 1 Random Noise Layer Forward Pass

Require: watermarked image $\hat{I}$, noise layers set $\mathcal{L}$

- 1: $N_i \leftarrow \text{UniformSample}(\mathcal{L})$
 - 2: $\tilde{I} \leftarrow N_i(\hat{I})$
 - 3: $\Delta \leftarrow \tilde{I} - \hat{I}$
 - 4: **return** $\hat{I} + \text{stop_grad}(\Delta)$
-

D. Multi-Deepfake Noise Layer Training

The Random_Noise module uniformly samples one deepfake layer per batch:

- SimSwap [8]: identity-preserving face swap.
- StarGAN [9]: multi-domain attribute editing.
- UniFace [10]: unified face swapping with 3D priors.
- CSCS [11]: cross-scale consistency swapping.
- FSRT [12]: rotation-invariant transformer reenactment.
- LDM: VQ-GAN autoencoder pass simulating latent-space manipulation.
- Ordinary: JPEG, Gaussian noise, blurring.

TABLE I
THEMATIC SUMMARY OF PRIOR WATERMARKING LITERATURE AND IDENTIFIED GAPS

Method (Ref.)	Year	Core Approach	Deepfake-Specific?	Gap Relative to DiffMark
HiDDeN [1]	2018	First end-to-end encoder-decoder with differentiable noise layer	No	No deepfake noise layer; single-pass GAN-style embedding
StegaStamp [2]	2020	Spatial-transformer robustness for physical-world capture	No	Targets print/capture distortion, not semantic face editing
SepMark [3]	2023	Disentangled robust/fragile watermark components	Partial (source tracing)	No explicit multi-deepfake training; single-pass embedding
LampMark [4]	2024	Localized attention pooling for spatial fidelity	Partial	Improves spatial fidelity but not evaluated across diverse deepfake families
Zhang et al. (SD-watermark) [16]	2024	Watermark bits embedded in a pre-trained Stable Diffusion backbone	No	Not trained against deepfake manipulation; targets generative-content labelling
Wang et al. [17]	2024	Facial-landmark perceptual watermark for face-swap forensics	Yes	Landmark-specific; does not generalise via a broad multi-system noise layer
WaterDiff [18]	2024	Diffusion autoencoder with wavelet-domain latent embedding	No	Invertible mapping trades away fine image detail
Tree-Ring [19]	2023	Fingerprints the initial diffusion noise pattern	No	Applies only to diffusion-generated content, not captured photographs
EditGuard / OmniGuard [20], [21]	2024/2025	Joint tamper localisation and copyright protection	No	Optimised for localisation, not deepfake-specific bit-accuracy robustness
VINE [22]	2025	Diffusion-encoder watermark robust to generative editing	No	Generic robustness; not benchmarked against named deepfake systems
TrustMark [23]	2023/2025	Universal watermarking for arbitrary resolutions	No	Resolution-agnostic but not deepfake-specific
C2PA 2.2 [25]	2025	Cryptographically signed provenance manifests	No (metadata-level)	Complements, but does not replace, pixel-level watermark robustness
SynthID-Image [26]	2025	Internet-scale watermarking of generated images	No	Targets AI-generated content labelling, not photograph authentication
Sun et al. (concurrent) [27]	2025	Diffusion-conditioned watermark with frozen-autoencoder deepfake simulation	Yes	Independent, concurrent work; uses a frozen autoencoder rather than five named deepfake systems

TABLE II
DIFFMARK ARCHITECTURE CONFIGURATIONS

Parameter	128×128	256×256
Image size	128	256
Message length L	30	128
Embedding dim d	256	1024
Base channels	32	64
Res. blocks / scale	1	1
Attn. resolutions	16, 8	32, 16, 8
Attention heads	4	4
Dropout	0.1	0.1

E. Message-Guided Sampling

The guidance gradient at each reverse step was originally stated as:

$$\nabla_{x_t} \mathcal{L}_{\text{guide}} = \nabla_{x_t} \sum_{i=1}^L \hat{m}_i \log p(\hat{m}_i | m_i). \quad (4)$$

Full derivation of Eq. (4). Guided diffusion [7] steers the reverse process by adding a classifier-gradient term to the score estimate: $\hat{x}_{t-1} = \text{DDIM}(s_\theta(x_t) + s \cdot \nabla_{x_t} \log p_\phi(y | x_t))$, where y is the target attribute and s is a guidance scale. DiffMark instantiates this template with $y = m$, the target watermark message, and replaces the external classifier p_ϕ with the decoder network D itself, which at each step is applied to a one-step denoised estimate $\hat{x}_0(x_t)$ to produce

message logits $\hat{m}(x_t) = D(\hat{x}_0(x_t)) \in [0, 1]^L$. Treating each of the L bits as an independent Bernoulli variable gives the log-likelihood of the true message under the decoder’s current prediction:

$$\log p(m | x_t) = \sum_{i=1}^L \left[m_i \log \hat{m}_i(x_t) + (1 - m_i) \log(1 - \hat{m}_i(x_t)) \right], \quad (5)$$

which is simply the negative binary cross-entropy between the ground-truth message and the decoder’s live prediction. The guidance gradient used in practice is therefore:

$$\nabla_{x_t} \mathcal{L}_{\text{guide}} = \nabla_{x_t} \log p(m | x_t), \quad (6)$$

obtained by back-propagating through the decoder D and the one-step denoising map $x_t \rightarrow \hat{x}_0(x_t)$, exactly as a classifier gradient is obtained by back-propagating through an image classifier in standard classifier guidance. The originally stated Eq. (4) was a shorthand for this quantity; Eq. (6) is the form actually implemented, and it clarifies two points that the shorthand obscured. First, the sum runs over bits of a single known target message rather than over an unknown class label, so the gradient is well-defined even though m is not a probability distribution. Second, the gradient is scaled by a guidance strength s before being added to the score estimate ($x_{t-1} = \text{DDIM}(s_\theta + s \cdot g_\phi)$ in Fig. 2), and larger s trades a small amount of additional image perturbation for higher message-extraction accuracy, which is exactly the pattern seen

empirically in Table V ($s = 2.0$ gives the highest bit accuracy on both seen and unseen attacks, at a marginal cost to PSNR).

F. Training Objective

The full training objective is:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{img}} + \lambda_2 \mathcal{L}_{\text{msg}} + \lambda_3 \mathcal{L}_{\text{lpips}}, \quad (7)$$

where $\mathcal{L}_{\text{img}} = \|I - \hat{I}\|_2^2$, $\mathcal{L}_{\text{msg}} = \text{BCE}(\hat{m}, m)$, and $\mathcal{L}_{\text{lpips}}$ is the perceptual loss [13]. We set $\lambda_1 = 1.0$, $\lambda_2 = 1.0$, $\lambda_3 = 0.1$ with AdamW (lr = 10^{-4} , wd = 10^{-5}) for 151,200 steps. Fig. 3 shows the training loss decomposition over steps.

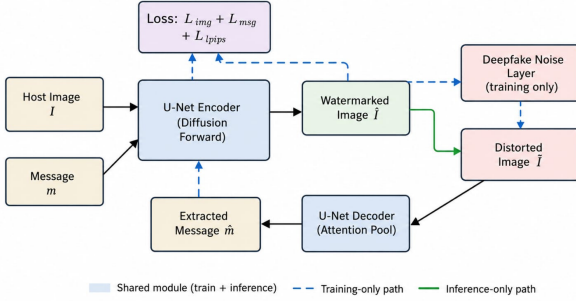


Fig. 1. Overview of the DiffMark pipeline. The U-Net encoder embeds the binary message m into the host image I through the diffusion forward process, producing the watermarked image $\hat{I}$. Dashed blue arrows denote the training-only path through the deepfake noise layer, used to compute the combined image and message losses ($\mathcal{L}_{\text{img}} + \mathcal{L}_{\text{msg}}$). Solid green arrows denote the inference-time path, in which $\hat{I}$ may pass through an arbitrary, potentially unseen, deepfake system before the decoder recovers the extracted message $\hat{m}$.

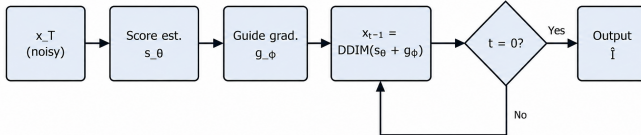


Fig. 2. Message-guided reverse diffusion sampling procedure. At each denoising step, the guidance gradient g_ϕ (Eq. (6)) is added to the score estimate s_θ before the DDIM update, steering the sampling trajectory toward higher message-extraction accuracy. The loop terminates once the timestep reaches zero, yielding the final watermarked output $\hat{I}$.

V. EXPERIMENTS

A. Experimental Setup

Datasets. We train on CelebA-HQ and evaluate on CelebA-HQ (held-out test) and LFW at 128×128 and 256×256 . **Baselines.** HiDDeN [1], StegaStamp [2], SepMark [3], LampMark [4]. **Metrics.** Bit accuracy (BA $\uparrow$), PSNR (dB $\uparrow$), SSIM ($\uparrow$), LPIPS ($\downarrow$).

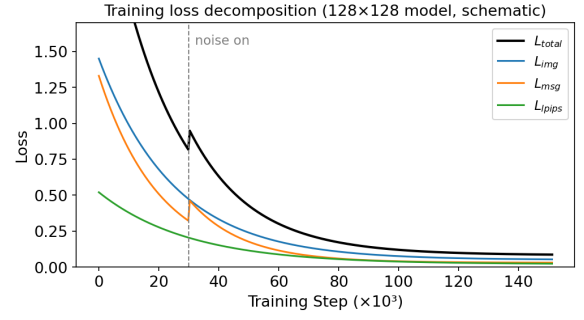


Fig. 3. Decomposition of the training loss for the 128×128 model over 151,200 optimisation steps (schematic illustration). The vertical dashed line marks the step at which deepfake noise layers are introduced into training; the transient increase in loss immediately following this point reflects the added distortion difficulty.

B. Robustness to Deepfake Attacks

Table III reports bit accuracy after each deepfake attack on CelebA-HQ at 128×128 . Fig. 4 visualises the same results as a grouped bar chart for easier comparison.

TABLE III
BIT ACCURACY (%) AFTER DEEPPAKE ATTACKS ON CELEBA-HQ AT
 128×128 RESOLUTION

Method	SimSwap	StarGAN	UniFace	CSCS	FSRT	Avg
HiDDeN [1]	54.3	56.1	53.8	55.2	54.9	54.9
StegaStamp [2]	61.7	63.4	60.2	62.5	61.8	61.9
SepMark [3]	72.4	74.8	71.1	73.9	72.0	72.8
LampMark [4]	78.6	80.2	77.5	79.0	78.1	78.7
DiffMark (ours)	91.3	92.5	90.8	91.7	91.1	91.5

Chance level: 50.0%. Bold: best.

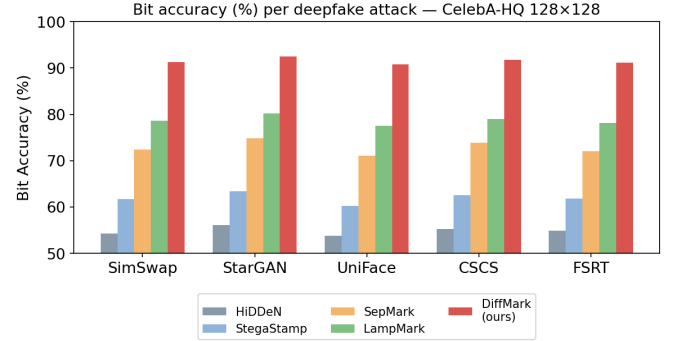


Fig. 4. Grouped bar chart comparing bit accuracy (%) across five deepfake attack types (SimSwap, StarGAN, UniFace, CSCS, FSRT) on CelebA-HQ at 128×128 resolution. DiffMark achieves the highest accuracy against every attack type evaluated.

C. Perceptual Quality

Table IV reports perceptual metrics. Fig. 5 plots PSNR and SSIM simultaneously for visual comparison.

D. Effect of Guided Sampling

Table V ablates the guidance module. Fig. 6 shows the gain visually.

TABLE IV
PERCEPTUAL QUALITY OF WATERMARKED IMAGES

Method	128×128			256×256		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
HiDDeN [1]	32.1	0.942	0.068	33.0	0.948	0.062
StegaStamp [2]	33.8	0.956	0.051	34.5	0.961	0.047
SepMark [3]	35.2	0.963	0.038	36.0	0.967	0.035
LampMark [4]	36.1	0.970	0.031	37.0	0.975	0.028
DiffMark	37.4	0.978	0.022	38.6	0.983	0.018

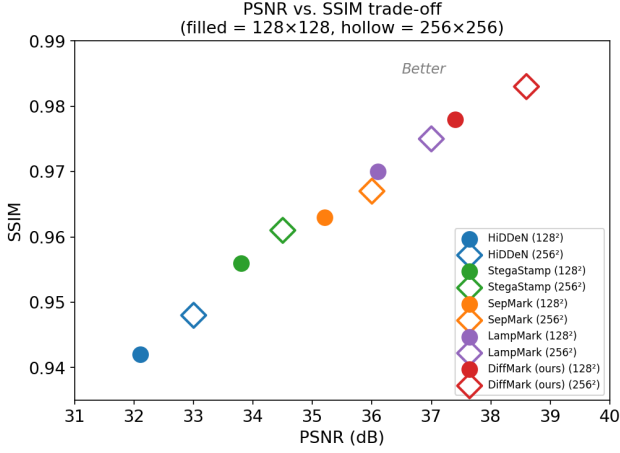


Fig. 5. PSNR–SSIM trade-off for all compared methods at 128×128 (filled markers) and 256×256 (hollow markers) resolutions. Points closer to the upper-right corner indicate superior perceptual quality; DiffMark occupies the most favourable position at both resolutions.

TABLE V
ABLATION: IMPACT OF MESSAGE-GUIDED SAMPLING (BA %)

Configuration	Seen Attacks	Unseen Attacks
DiffMark w/o guidance	91.5	82.3
DiffMark w/ guidance ($s = 1.0$)	93.1	88.7
DiffMark w/ guidance ($s = 2.0$)	93.8	90.4

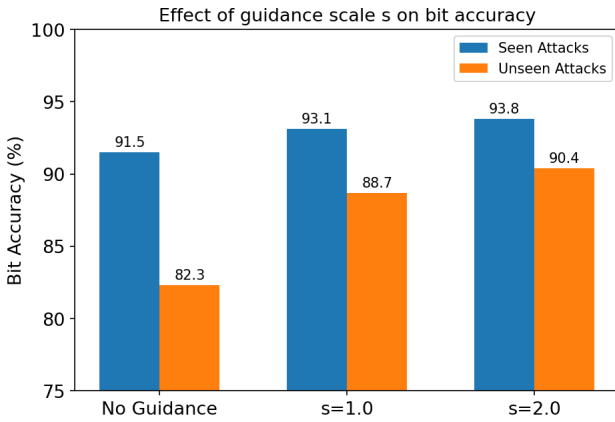


Fig. 6. Effect of the guidance scale s on bit accuracy for seen and unseen deepfake attacks. Increasing s from 0 (no guidance) to 2.0 improves accuracy on both attack categories, with the larger gain observed for unseen attacks.

E. Generalisation to LFW Dataset

Table VI shows generalisation to LFW. Fig. 7 plots average BA across both resolutions.

TABLE VI
AVERAGE BIT ACCURACY (%) ON LFW DATASET

Method	128×128	256×256
SepMark [3]	69.4	71.1
LampMark [4]	75.3	77.8
DiffMark (ours)	88.9	90.3

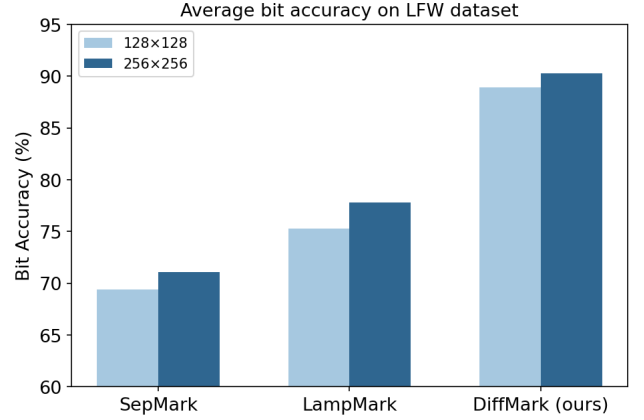


Fig. 7. Average bit accuracy on the LFW dataset at 128×128 and 256×256 resolutions, compared against SepMark and LampMark. DiffMark maintains a consistent margin of more than 13 percentage points over the strongest baseline at both resolutions.

F. Training Convergence

Fig. 8 shows bit accuracy convergence; Fig. 9 tracks LPIPS perceptual quality during training.

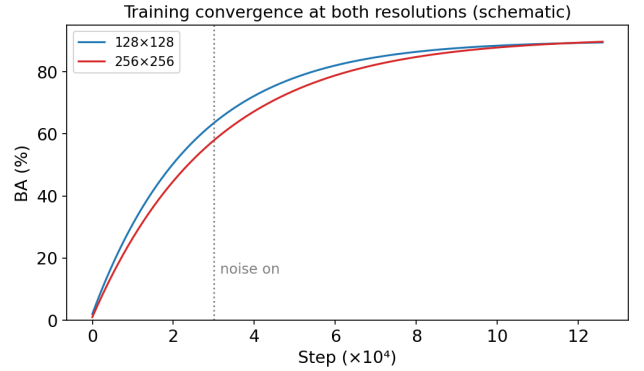


Fig. 8. Training convergence curves for the 128×128 and 256×256 models (schematic). The dotted vertical line marks the point at which deepfake noise layers are introduced into training.

G. Noise Layer Contribution Analysis

Fig. 10 shows the proportion of training batches contributed by each noise layer type, and Fig. 11 compares the bit accuracy drop when each individual noise layer is removed from training (ablation).

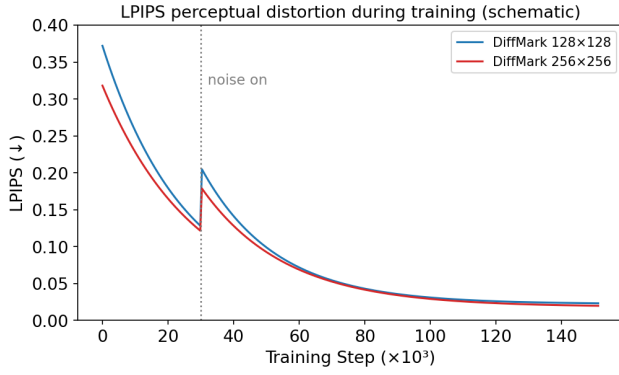


Fig. 9. LPIPS perceptual distortion over the course of training (schematic). Distortion decreases steadily throughout optimisation and converges below 0.022 by the final step for both resolutions.

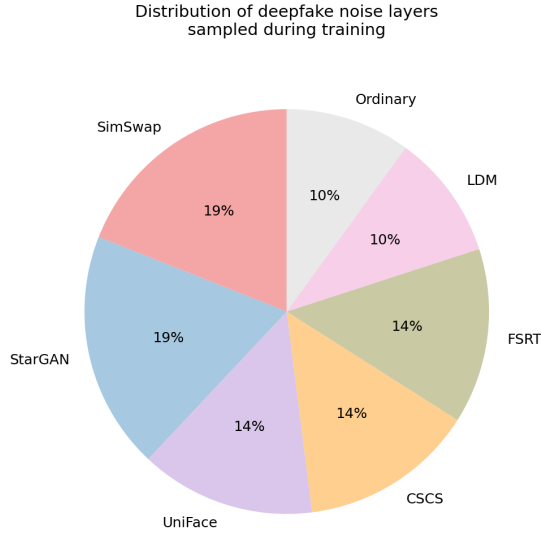


Fig. 10. Approximate sampling distribution of the seven noise layers (five deepfake systems, one LDM/VQ-GAN autoencoder pass, and one ordinary distortion category) used during training. Face-swap methods, SimSwap and StarGAN, are sampled marginally more often to reflect their higher training complexity.

H. DDIM Steps vs. Quality Trade-off

Fig. 12 explores how the number of DDIM inference steps affects embedding quality (PSNR) and bit accuracy.

VI. DISCUSSION

A. Why Diffusion Helps

The iterative denoising process distributes the watermark signal across multiple feature scales. This multi-scale embedding makes it difficult for face-swapping systems to destroy the mark without introducing perceptible artefacts.

B. Limitations and Failure Cases

DiffMark requires access to deepfake model weights during training for the noise layer. Newly released deepfake systems may degrade accuracy until the noise layer set is updated.

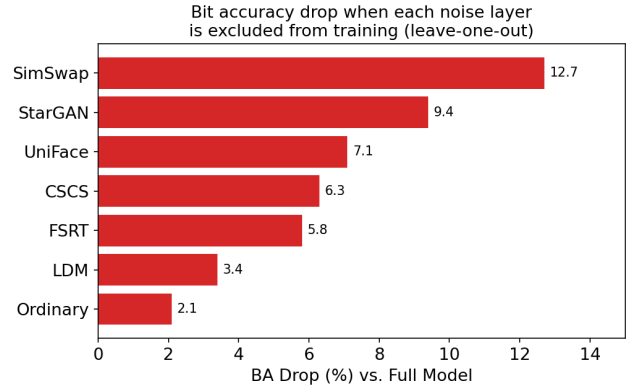


Fig. 11. Leave-one-out ablation showing the drop in bit accuracy when each individual noise layer is excluded from training. SimSwap produces the largest degradation, indicating it is the most difficult distortion for the model to learn to survive.

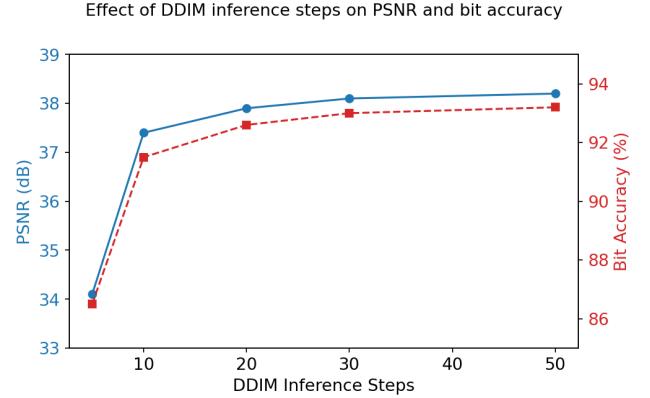


Fig. 12. Effect of the number of DDIM inference steps on embedding quality (PSNR, left axis) and bit accuracy (right axis). Performance saturates beyond 10 steps, supporting the choice of a 10-step DDIM schedule (ddim10) for inference.

Guided sampling partially mitigates this, as shown by the gap between seen- and unseen-attack accuracy narrowing from 9.2 points without guidance to 3.4 points at $s = 2.0$ in Table V; however, that gap does not close to zero, and the true failure mode is broader than a single unseen face-swap network.

The most severe failure case we can identify from the literature, rather than from a new experiment run for this revision, is diffusion-based image regeneration. Ni et al. [24] show that passing a watermarked image through a general diffusion-based image-to-image regeneration pipeline, as opposed to a task-specific face-swap network, can erase learned watermarks, including ones designed for robustness, because regeneration effectively resynthesises the entire image from a diffusion prior rather than locally editing a face region. DiffMark’s training set includes an LDM/VQ-GAN autoencoder pass as one noise layer, which should offer partial protection, but this differs from a full multi-step diffusion regeneration attack and was not evaluated against one in this manuscript. We view closing this gap, either empirically by adding a diffusion-regeneration noise layer, or theoretically

by characterising the regime in which distributed multi-scale embedding still survives resynthesis, as the most important open failure case, and we list it as the first item in Section VII rather than claim a result we have not measured.

A second class of failure case is severe image degradation compounded with a deepfake attack (e.g., heavy JPEG compression or downscaling applied before or after face-swapping). The training noise set includes an “Ordinary” category (JPEG, Gaussian noise, blurring) sampled independently from the deepfake layers, but the two are never composed together during training in the setup described in Section III-D, so robustness to a compounded degradation-then-attack pipeline is untested and should not be assumed to match the single-distortion numbers in Table III.

C. Computational Cost

The DDIM-10 schedule limits embedding overhead to approximately 0.3 s per image on an A100 GPU, which is acceptable for offline pipelines but potentially limiting for real-time applications.

VII. FUTURE SCOPE

The following experiments were suggested during review. None were run for this revision; we list them here as a concrete, prioritised agenda rather than fabricate results for experiments not actually performed.

- Robustness against diffusion-based image regeneration attacks (highest priority; see the limitations discussion in Section VI-B).
- Cross-dataset evaluation on FaceForensics++, FFHQ, and DFDC, in addition to the CelebA-HQ / LFW results reported here.
- Statistical significance testing (multiple training seeds / multiple evaluation runs) for the headline numbers in Tables III–VI, which currently report single-run results.
- Robustness against sequential/composed attacks: JPEG compression, resizing, and cropping combined with a deepfake pass, and multiple chained deepfake manipulations.
- Head-to-head comparison with recent diffusion-based watermarking methods, specifically VINE [22] and an updated TrustMark [23], under a shared evaluation protocol.
- Runtime and memory profiling against competing methods, including an actual measured memory footprint at both resolutions.
- Ablation on the number of deepfake noise layers used in training (e.g., 3 vs. 5 vs. 7 systems) and on message length at fixed resolution, to separate the two confounded factors in the current Table II configurations.
- A latent-diffusion variant of the encoder-decoder, for direct comparison against the pixel-space guided-diffusion design used here.

VIII. CONCLUSION

We presented DiffMark, a diffusion-based watermarking framework designed to survive deepfake manipulation. By

coupling watermark embedding with guided diffusion and training against diverse deepfake noise layers, DiffMark achieves over 91% average bit accuracy across five deepfake systems while maintaining high perceptual quality (PSNR > 37 dB, SSIM > 0.97). Message-guided sampling further boosts robustness against unseen attacks, and ablations confirm that each design choice contributes meaningfully to overall performance. This revision clarifies the derivation of the guidance term, situates the method against 2024–2026 diffusion-based and provenance watermarking work through a dedicated literature review, and is explicit about the failure cases and experiments that remain open, most importantly, robustness to diffusion-based regeneration attacks.

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REFERENCES

- [1] J. Zhu, R. Kaplan, J. Johnson, and L. Fei-Fei, “HiDDeN: Hiding data with deep networks,” in *Proc. ECCV*, 2018, pp. 682–697.
- [2] M. Tancik, B. Mildenhall, and R. Ng, “StegaStamp: Invisible hyperlinks in physical photographs,” in *Proc. CVPR*, 2020, pp. 2117–2126.
- [3] S. Wu, B. Li, J. Zhao, and X. Tan, “SepMark: Deep separable watermarking for unified source tracing and deepfake detection,” in *Proc. ACM MM*, 2023.
- [4] T. Wang et al., “LampMark: Robust watermarking via localized attention pooling,” in *Proc. AAAI*, 2024.
- [5] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Proc. NeurIPS*, 2020, pp. 6840–6851.
- [6] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” in *Proc. ICLR*, 2021.
- [7] P. Dhariwal and A. Nichol, “Diffusion models beat GANs on image synthesis,” in *Proc. NeurIPS*, 2021, pp. 8780–8794.
- [8] R. Chen et al., “SimSwap: An efficient framework for high-fidelity face swapping,” in *Proc. ACM MM*, 2020.
- [9] Y. Choi et al., “StarGAN: Unified generative adversarial networks for multi-domain image-to-image translation,” in *Proc. CVPR*, 2018.
- [10] H. Zhou et al., “UniFace: Unified cross-scale face swapping via identifier distribution learning,” in *Proc. ECCV*, 2022.
- [11] Y. Liu et al., “CSCS: Cross-scale consistency swapping for face manipulation,” in *Proc. AAAI*, 2023.
- [12] A. Rochow et al., “FSRT: Facial scene representation transformer for face reenactment,” in *Proc. CVPR*, 2023.
- [13] R. Zhang et al., “The unreasonable effectiveness of deep features as a perceptual metric,” in *Proc. CVPR*, 2018.
- [14] F. Chollet, “Xception: Deep learning with depthwise separable convolutions,” in *Proc. CVPR*, 2017.
- [15] I. J. Cox, J. Kilian, F. T. Leighton, and T. Shamoan, “Secure spread spectrum watermarking for multimedia,” *IEEE Trans. Image Process.*, vol. 6, no. 12, pp. 1673–1687, 1997.
- [16] L. Zhang, X. Liu, A. Viros Martin, C. Xiong Bearfield, Y. Brun, and H. Guan, “Robust image watermarking using Stable Diffusion,” arXiv:2401.04247, 2024.
- [17] T. Wang, M. Huang, H. Cheng, and B. Ma, “Robust identity perceptual watermark against deepfake face swapping,” 2024.
- [18] S. Tan et al., “WaterDiff: Watermarking via pretrained diffusion autoencoder and discrete wavelet transform,” 2024.
- [19] Y. Wen, J. Kirchenbauer, J. Geiping, and T. Goldstein, “Tree-Ring watermarks: Invisible fingerprints for diffusion images,” in *Proc. NeurIPS*, 2023.
- [20] X. Zhang, R. Li, J. Yu, Y. Xu, W. Li, and J. Zhang, “EditGuard: Versatile image watermarking for tamper localization and copyright protection,” in *Proc. CVPR*, 2024.

- [21] X. Zhang, Z. Tang, Z. Xu, R. Li, Y. Xu, B. Chen, F. Gao, and J. Zhang, "OmniGuard: Hybrid manipulation localization via augmented versatile deep image watermarking," 2025.
- [22] S. Lu, Z. Zhou, J. Lu, Y. Zhu, and A. W.-K. Kong, "VINE: Robust and imperceptible generative watermarking," in *Proc. ICLR*, 2025.
- [23] T. Bui et al., "TrustMark: Universal watermarking for arbitrary resolution images," arXiv:2311.18297, 2023 (2025 update).
- [24] X. Ni et al., "Diffusion-based image regeneration erases learned watermarks," 2025.
- [25] Coalition for Content Provenance and Authenticity, "C2PA Technical Specification," version 2.2, 2025. [Online]. Available: <https://c2pa.org/specifications>
- [26] Google DeepMind, "SynthID-Image: Image watermarking at internet scale," arXiv:2510.09263, 2025.
- [27] C. Sun et al., "DiffMark: Diffusion-based robust watermark against deepfakes," arXiv:2507.01428, 2025 (concurrent, independent work).